



# **CHEM F111 : General Chemistry**

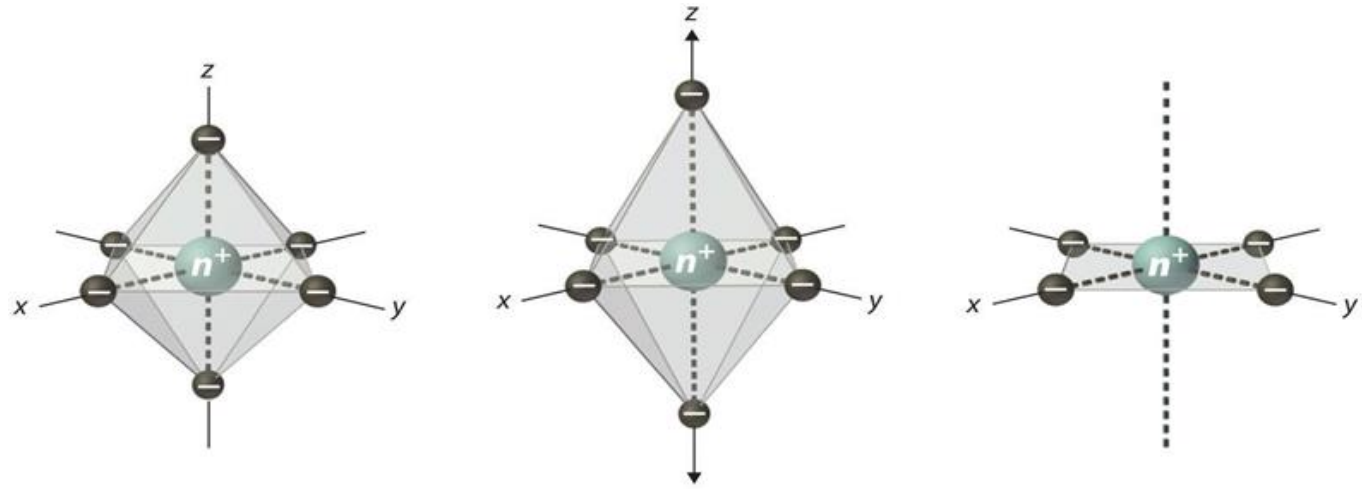
## **Lecture 22**

### **Inorganic Chemistry**

# Discussed topics.....



- ✓ Square planar energy level



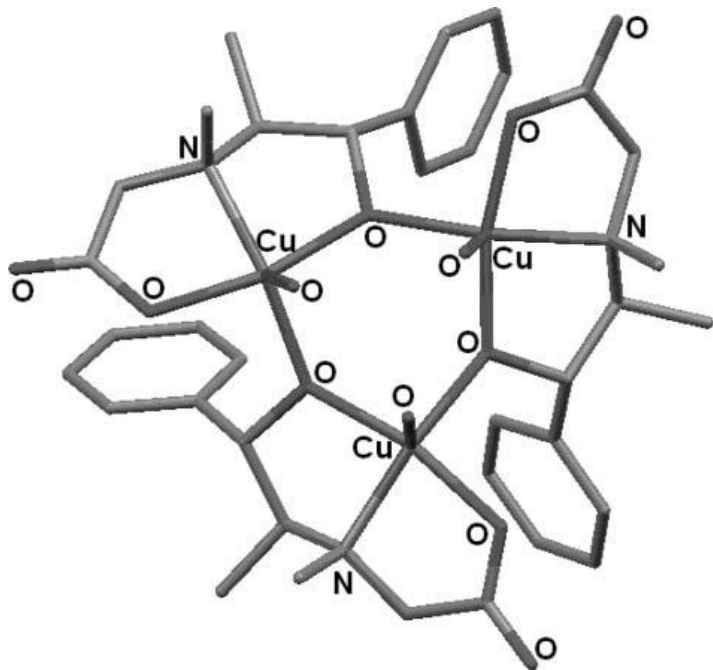
- ✓ Preferred geometry of Cu(II) Complex

# Representative Cu complexes

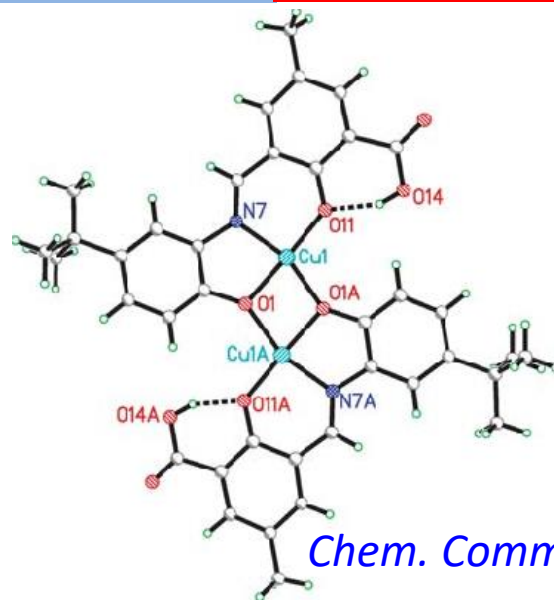
innovate

achieve

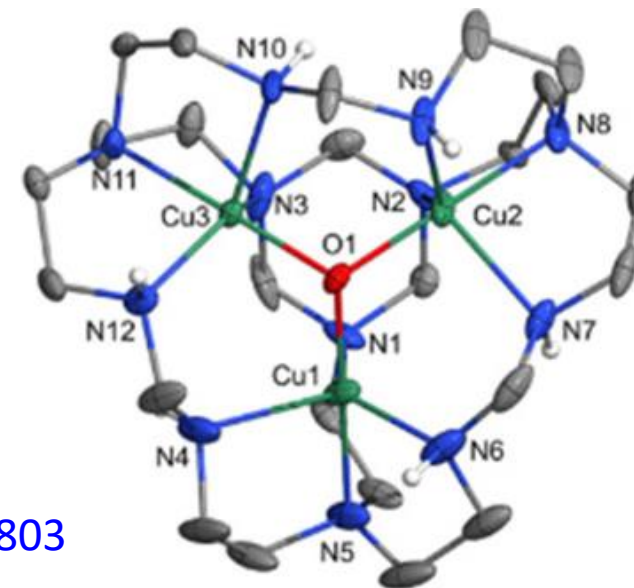
lead



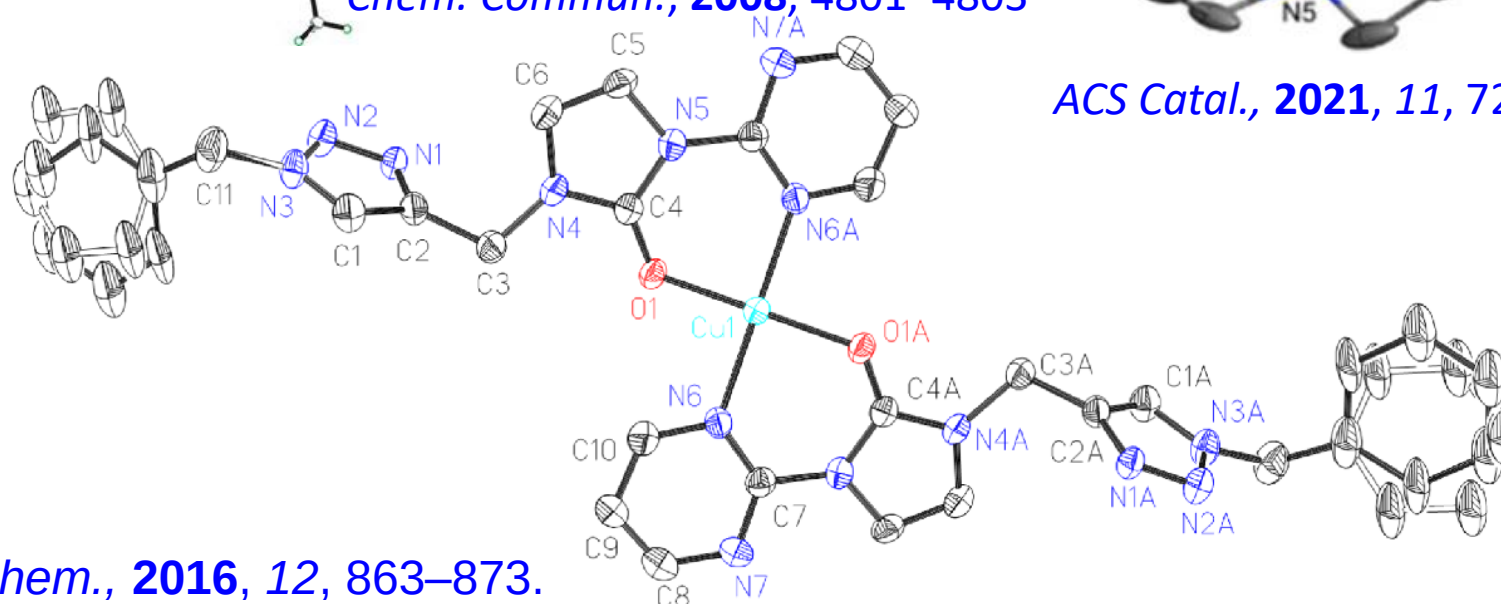
*Dalton Trans.*, **2007**, 4185–4196



*Chem. Commun.*, **2008**, 4801–4803



*ACS Catal.*, **2021**, 11, 7223–7240



*Beilstein J. Org. Chem.*, **2016**, 12, 863–873.

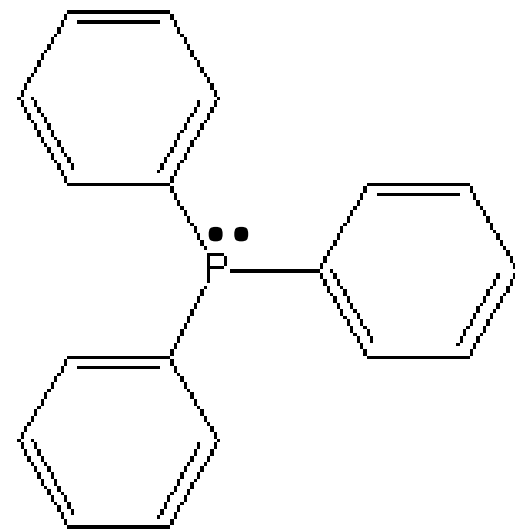
# Tetrahedral Complexes

Octahedral complexes are preferred for

- 6 bond energies compared to 4 in tetrahedral
- greater CFSE

• Then why Tetrahedral complexes are formed?

❖ When the ligands are large and bulky and cause crowding; In such cases the steric hindrance is of higher energy in octahedral than in tetrahedral.  
e. g. triphenyl phosphine



# Tetrahedral Complexes

innovate

achieve

lead

❖ Where attainment of regular shape is important.

e.g. compare  $d^7$

In a strong octahedral field  $(t_{2g})^6 (e_g)^1$

Asymmetry in  $e_g \rightarrow$  distorted

In a tetrahedral field  $e^4 t_2^3$

symmetry in  $e$  and  $t_2 \rightarrow$  Regular

$[\text{CoCl}_4]^{2-}$  is regular tetrahedron

$d^0$ ,  $d^2$ ,  $d^5$ ,  $d^7$  and  $d^{10}$  configurations of tetrahedral geometry are regular.

# Tetrahedral Complexes



❖ When the CFSE is comparable both in octahedral and tetrahedral geometries.

$d^0$ ,  $d^5$ ,  $d^{10}$  no CFSE in oct. and tet.

$d^1$  and  $d^6$  (weak) has

Oct and Tet CFSE difference

$$= -0.4 \Delta_o - (-0.6 \Delta_t) = (-0.4 - (-0.6 \times (4/9))) \Delta_o$$

$$= (-0.4 - (-0.27)) \Delta_o = -0.13 \Delta_o$$

Similarly for  $d^2$  and  $d^7$  cases the difference is

Only  $-0.17 \Delta_o$

# Tetrahedral Complexes

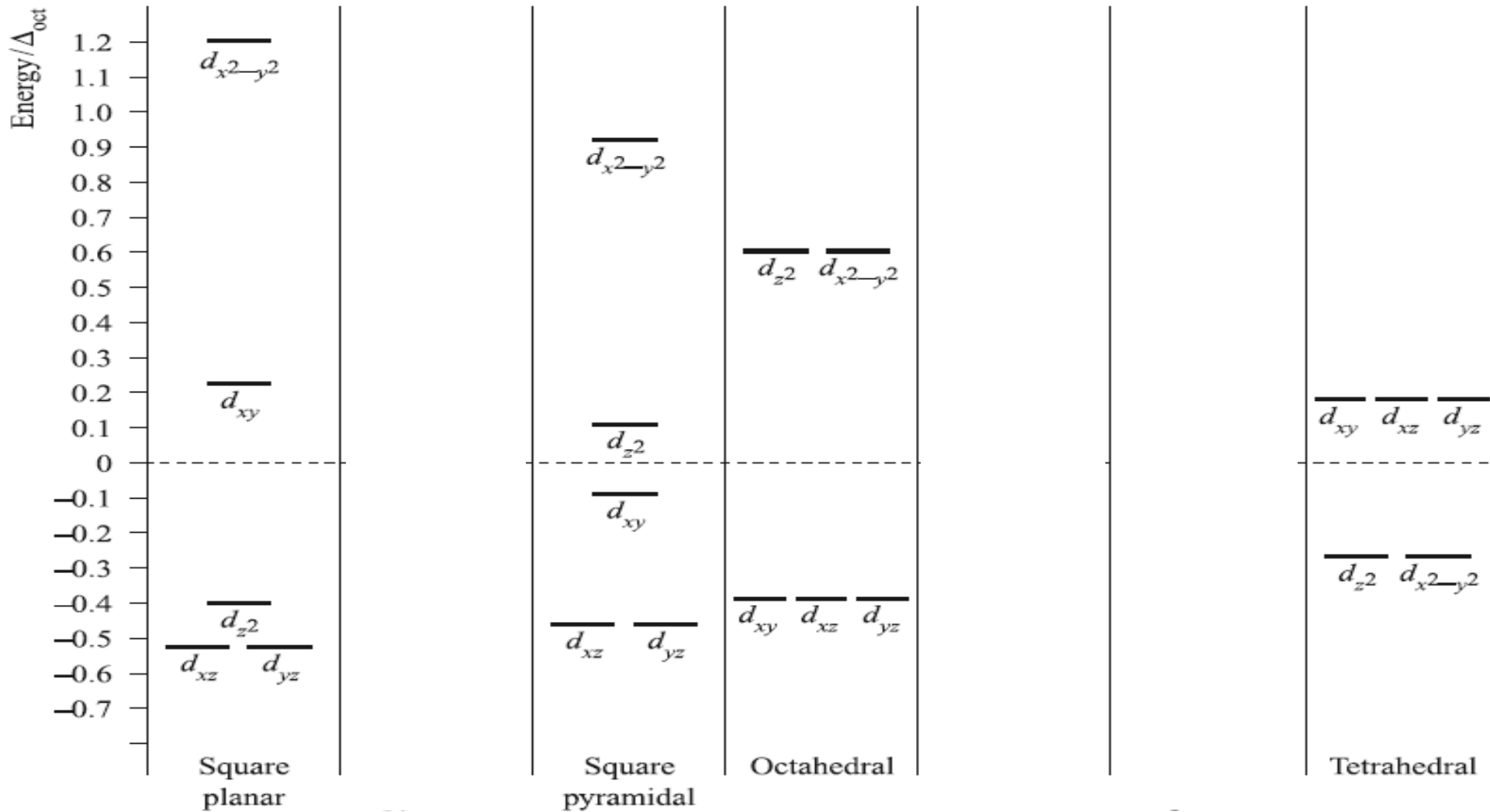


❖ When the ligands are weak then  $\Delta$  is less and CFSE is less in value.

If the charge on the metal ion is low (lower oxidation state) then also  $\Delta$  and CFSE are less in value.

That is why many transition metal chlorides, bromides and iodides form tetrahedral Complexes.

# Comparison of Crystal Field Splitting



Crystal field splitting diagrams for some common fields referred to a common Bary centre. Splittings are given with respect to  $\Delta_o$

# Jahn-Teller distortion in tetrahedral



Are Jahn-Teller distortions possible in tetrahedral environments? Do you think such distortions are likely to occur? Justify your answer.

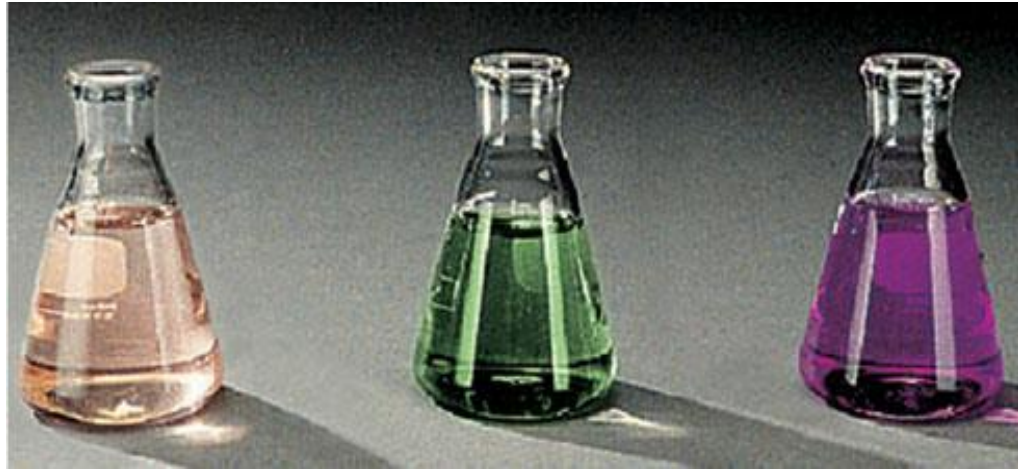
# Jahn-Teller distortion in tetrahedral



Are Jahn-Teller distortions possible in tetrahedral environments? Do you think such distortions are likely to occur? Justify your answer.

Jahn-Teller distortions are possible **whenever there is a partially filled degenerate set of energy levels**. This can of course occur here if either the  $e$  or  $t_2$  levels are only partially filled. However, since **neither set of orbitals is oriented directly towards the ligands**, the relative amount of **stabilization gained is smaller** than when the  $e_g$  set is partially occupied in an octahedron so such distortions are much less likely to occur in a tetrahedral environment

# Transition metal complexes

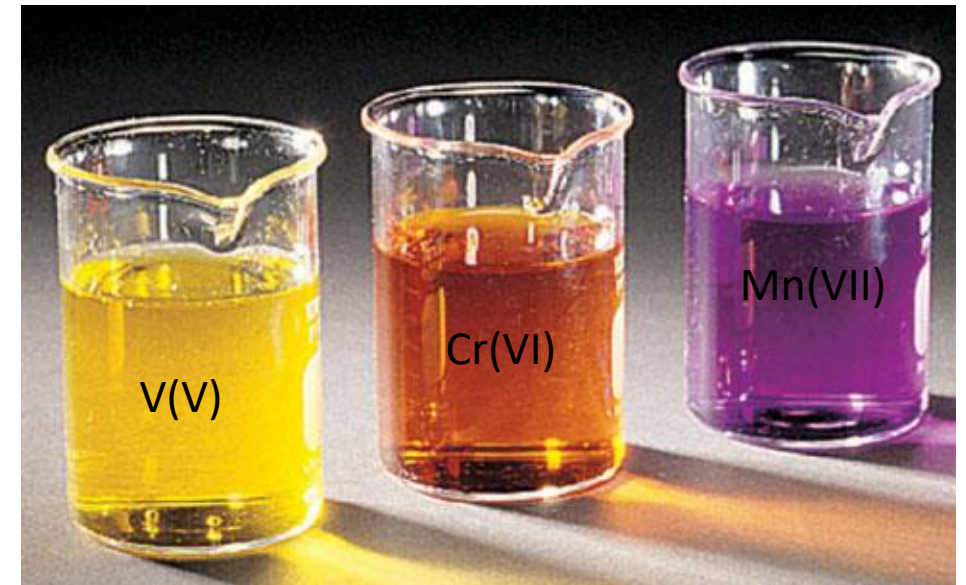
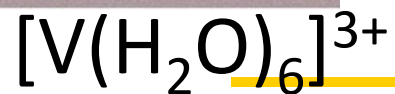
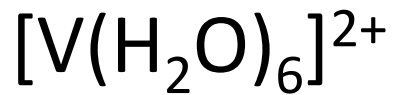


Mn(II)

Mn(VI)

Mn(VII)

One of the most characteristic chemical properties of these elements is the occurrence of *multiple oxidation states*.



V(V)

Cr(VI)

Mn(VII)

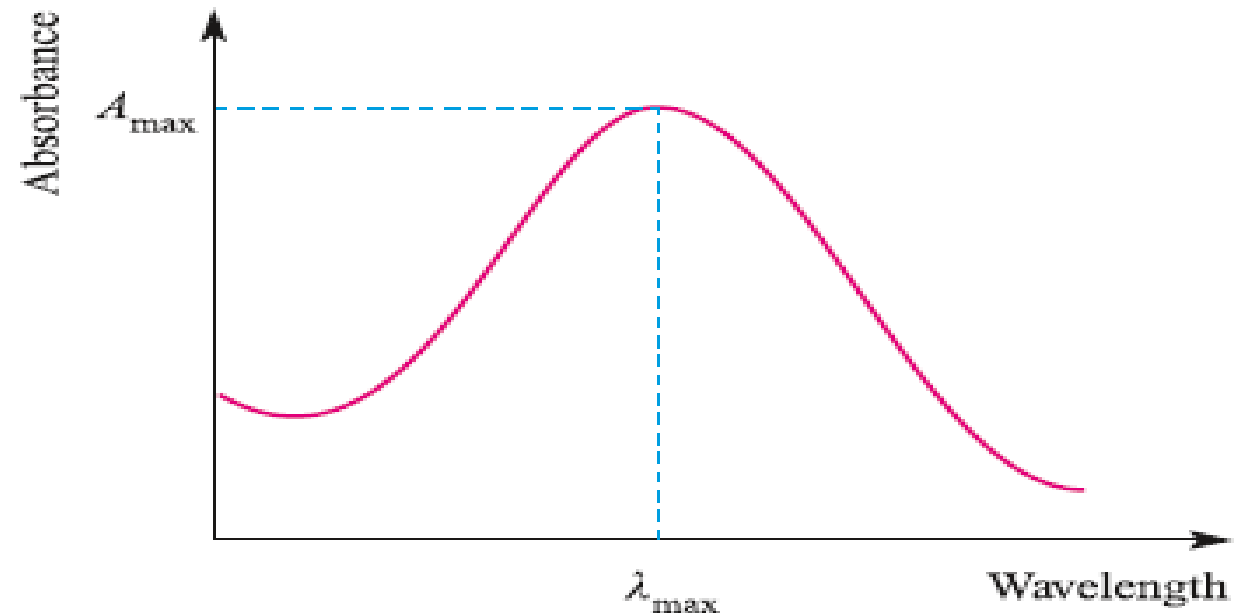
# Electronic spectroscopy of complexes



- Studies of electronic spectra of metal complexes provide information about structure and bonding, although interpretation of the spectra is not always straightforward.
- Spectra arise because of transition of electrons from one energy level to another by absorption of EM radiation in the UV-Visible region
- The spectrum of a colored solution may be measured quite easily in a spectrophotometer using the Lambert-Beer's Law

$$\log_{10}(I_0/I) = A$$

$$A = \epsilon \cdot c \cdot l$$



# Electronic spectroscopy of complexes

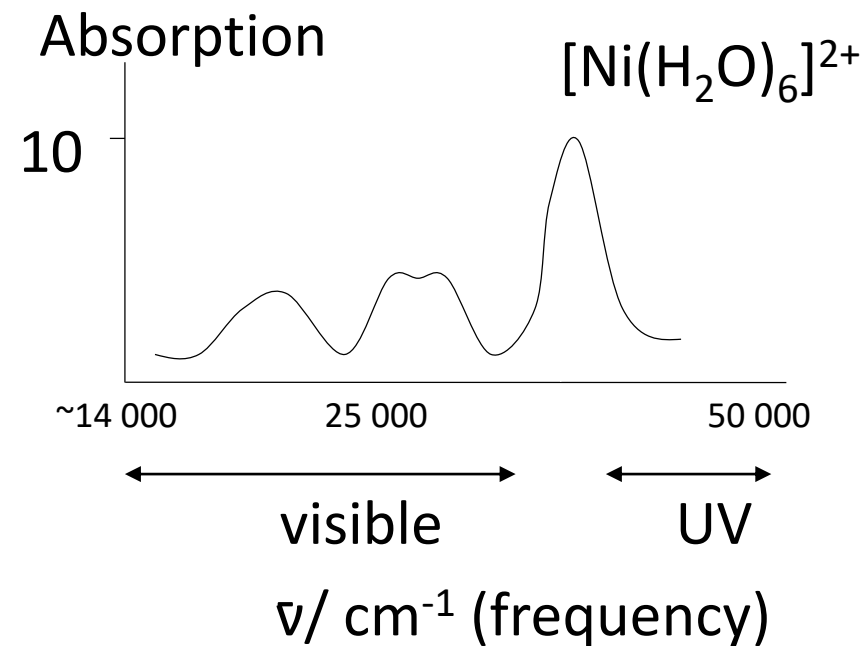
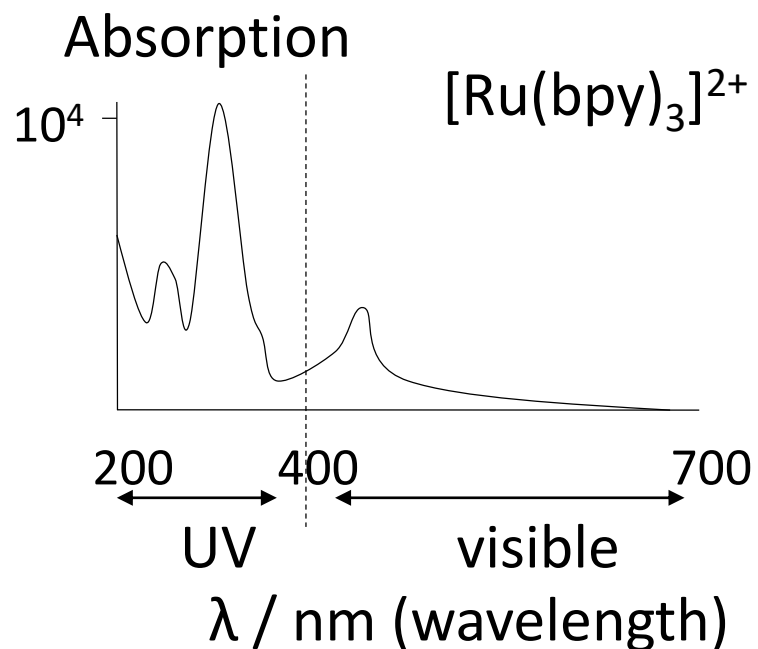


- Absorptions arise from transitions between electronic energy levels:
  - ✓ transitions between metal-centred orbitals possessing d-character ('d–d transition bands')
  - ✓ transitions between metal- and ligand-centred Mos, which transfer charge from metal to ligand or ligand to metal called charge transfer bands, (CT bands)
- An absorption band is characterized by both the wavelength ( $\lambda_{\text{max}}$ ) of the absorbed EM radiation and  $\epsilon_{\text{max}}$
- **An absorption spectrum** may be represented as a plot of **absorbance (A)** against **wavelength ( $\lambda$ )** or  **$\epsilon$  against wavenumber** , or **A against wavenumber** etc.

# Electronic spectroscopy of $O_h$ com.



Absorption of radiation leading to electronic transitions within a molecule or complex



- UV = higher energy transitions - between ligand orbitals
- Visible = lower energy transitions - between d-orbitals of transition metals
- between metal and ligand orbitals

# Electronic spectroscopy of $O_h$ complex



Absorption maxima in a visible spectrum have **three important characteristics**

1. **number** (how many)

This depends on the electron configuration of the metal centre

2. **position** (what wavelength/energy)

This depends on the ligand field splitting parameter,  $\Delta_o$  or  $\Delta_t$  and on the degree of inter-electron repulsion

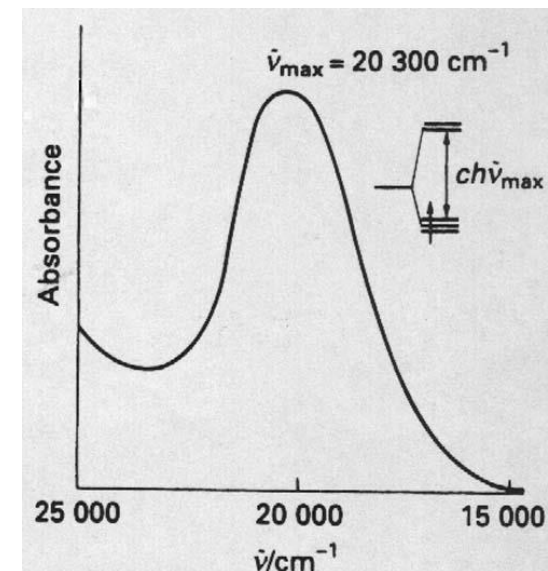
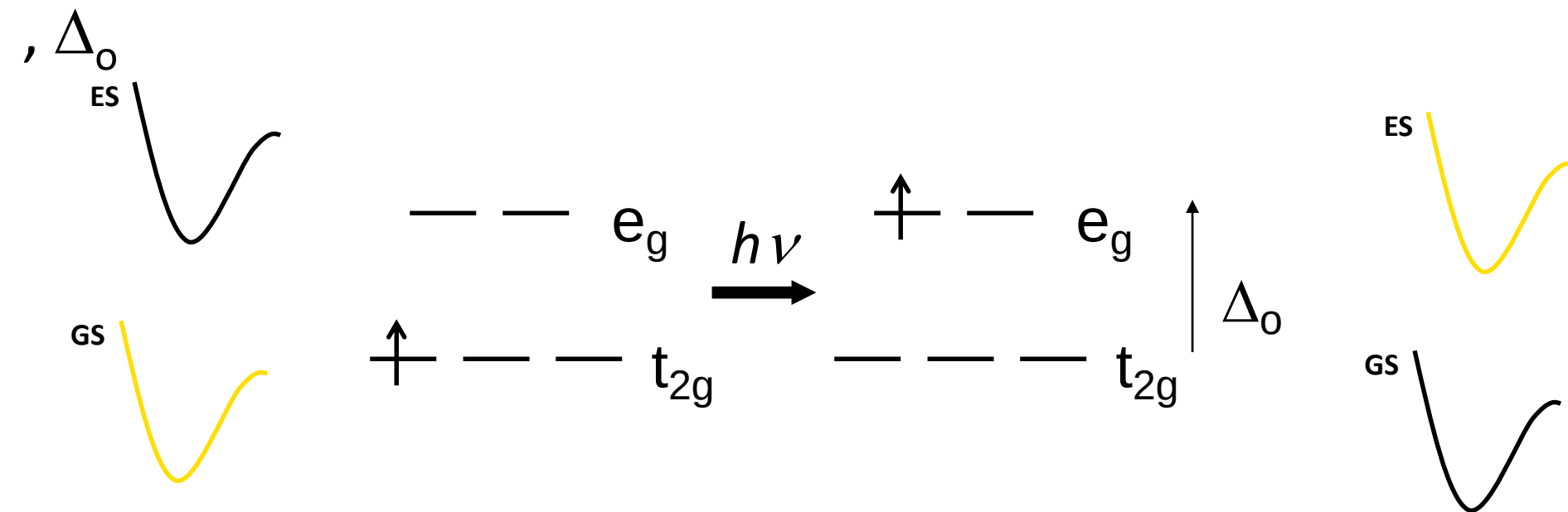
3. **intensity**

This depends on the "allowedness" of the transitions, which is described by two selection rules

# Electronic spectroscopy of Oh com.



The energy of the absorption by  $[\text{Ti}(\text{OH}_2)_6]^{3+}$  is the ligand-field splitting energy



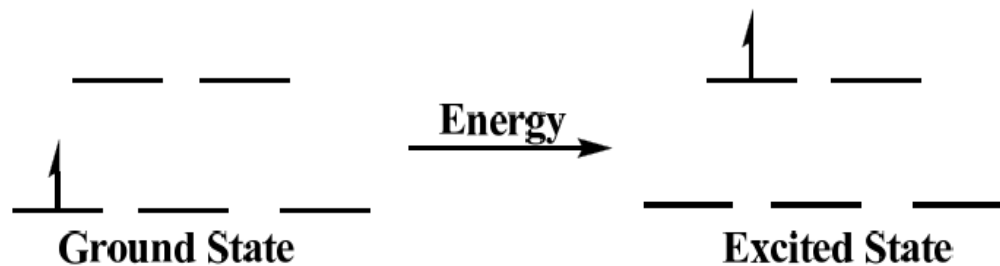
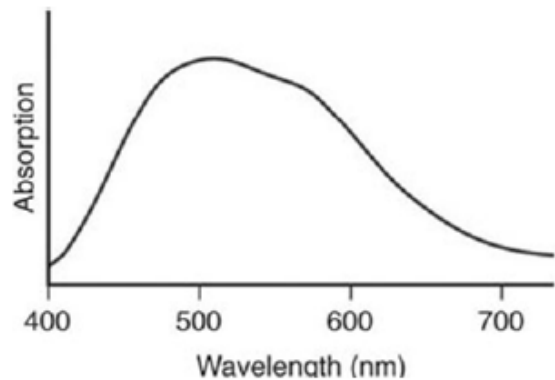
complex in electronic  
ground state (GS)

→ d-d transition →

complex in electronic  
excited state (ES)

$[\text{Ti}(\text{OH}_2)_6]^{3+}$   $\lambda_{\text{max}} = 510 \text{ nm}$   $\Delta_o$  is  $243 \text{ kJ mol}^{-1}$  or  $20\,300 \text{ cm}^{-1}$   
( $3d^1$  system)

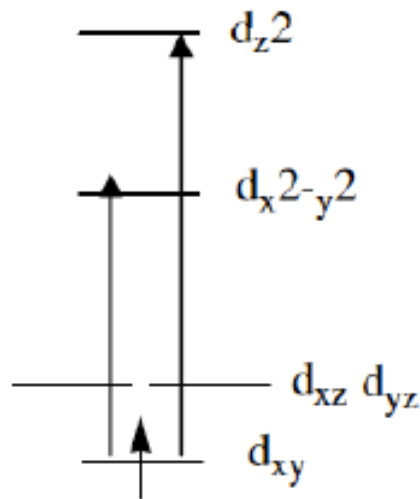
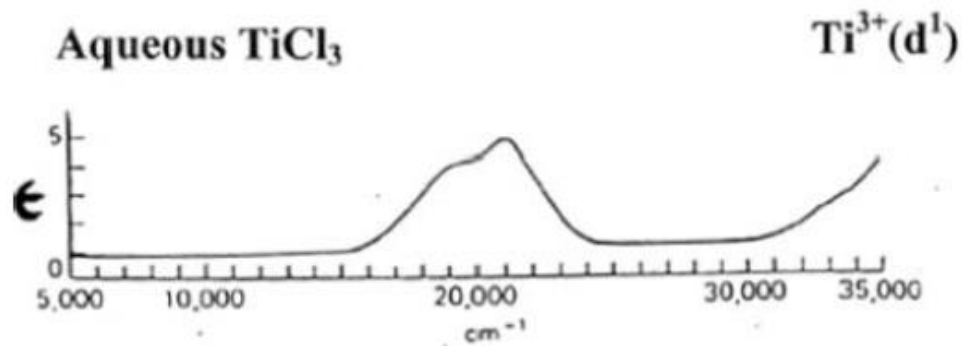
# Color of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ complex



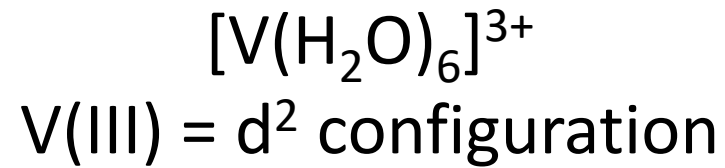
Absorption at 520 nm gives the complex its purple color



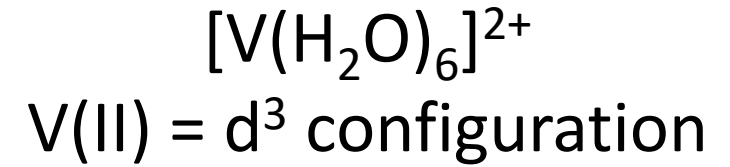
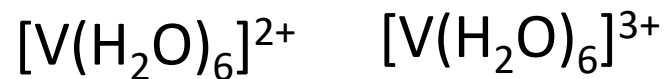
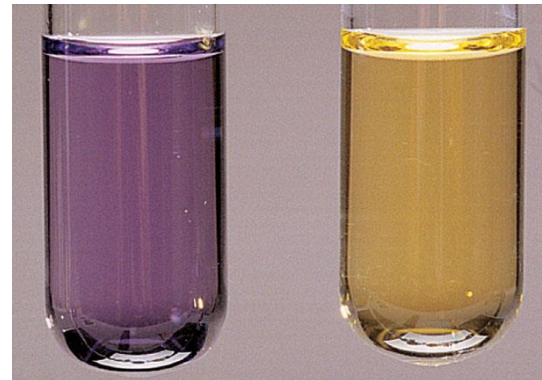
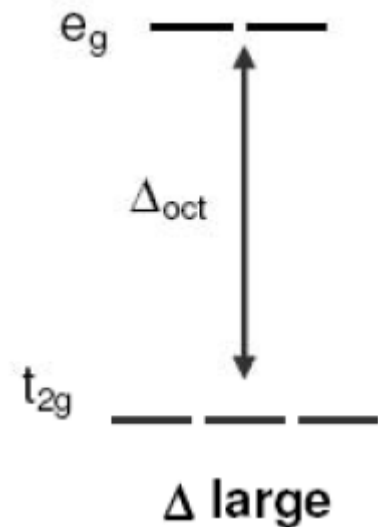
A more resolved absorption spectrum of the complex has a shoulder



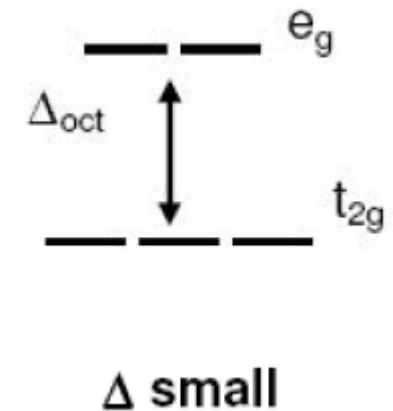
# Color and CFT



Violet light absorbed  
complex appears yellow



Yellow light absorbed  
complex appears violet



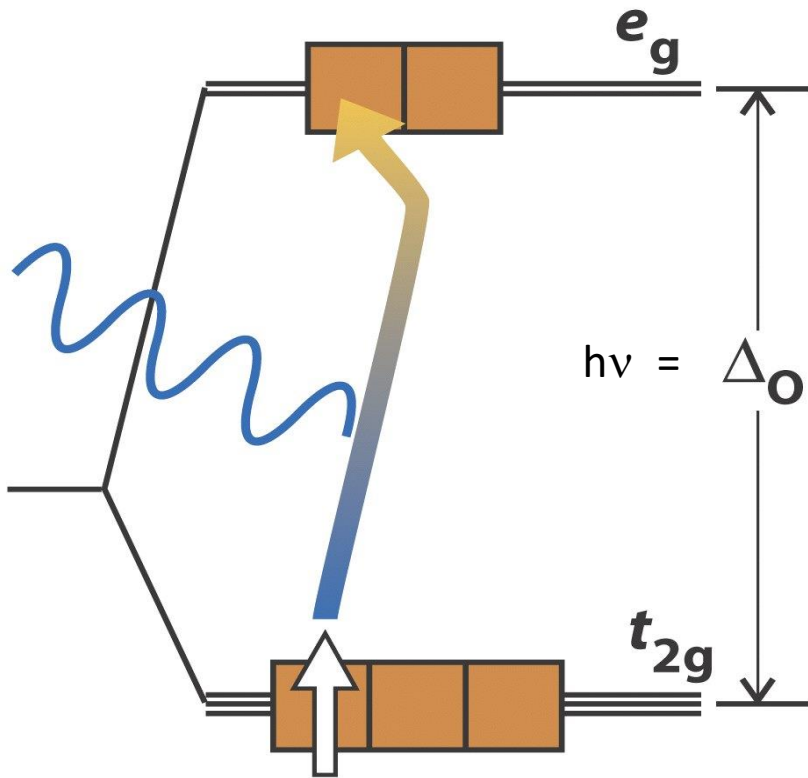
# Why different colors??



The color of the aqueous solution is due to the transition of electrons from one d level to another d level.....**why not only one color??**  
different colors are seen for different compounds because,

- ✓ The d orbitals are at different energy levels and hence there are different transition (3d, 4d, 5d)
- ✓ The splitting of d orbitals in different environments
- ✓ The splitting depends upon the geometry of the complex, the nature of the ligands, and the oxidation state of the central metal atom

# Electronic spectroscopy of $O_h$ com.



The energy gap between the  $e_g$  and  $t_{2g}$  orbitals,  $\Delta_o$ , (the crystal field splitting) equals the energy of a photon:

$$\Delta_o = h\nu = \Delta E$$

As  $\Delta_o$  varies,  $h\nu$  will also vary and the color of the compound will change

Absorption of a photon causes a jump from a  $t_{2g}$  to an  $e_g$  orbital

# Electronic spectroscopy of $O_h$ com.



$d^1$ ,  $d^4$ ,  $d^6$  and  $d^9$  usually have **one absorption**, though a side “hump”

